Flight Delay Propagation Analytics

An End-to-End Analysis of U.S. Flight Operations in 2025

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Stack: Python · Jupyter · PostgreSQL · SQL · Microsoft Power BI
Scope: 7,001,619 U.S. flight records · January–December 2025

Table of Contents

1. Executive Summary
2. Business Problem and Motivation
3. Project Objectives and Analytical Questions
4. Dataset Description and Scope
5. Technology Stack and Architecture
6. Data Ingestion, Cleaning, Transformation, and ETL
7. PostgreSQL Layer and Analytical Views
8. KPI Definitions and Calculation Methodology
9. Dashboard Design: The Four Pages
10. Key Findings and Operational Insights
11. Data-Quality Considerations and Limitations
12. Recommendations for Airline Operations Analysis
13. Conclusion
14. Reproducibility Notes

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1. Executive Summary

This project analyzes **7,001,619 U.S. flight records from 2025** to understand how delays arise, how often they are recovered, and how a delay on one flight relates to the departure performance of the next flight flown by the same aircraft. Data was processed with Python, stored and aggregated in PostgreSQL, and presented in a four-page interactive Power BI dashboard.

Headline results:

Theme	Result
Delay causes	Late-aircraft (43,862,971 min) and carrier (36,326,377 min) delays together make up **71.64%** of attributed delay minutes
Delay propagation	**21.20%** of 6,688,507 valid aircraft connections were followed by a delayed flight
Propagation after a delayed flight	In 2,469,518 connections following a previous flight with a recorded delay, **983,355 (≈39.82%)** next flights were delayed
Overall operations	Departure delay rate 21.45%, cancellation rate 1.47%, diversion rate 0.28%

The analysis shows a strong **association** between earlier-flight delays, short turnaround windows, and later-flight delays. It does **not** prove that one delay caused another; see Section 11.

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2. Business Problem and Motivation

Delays are costly for airlines (crew, fuel, connections, compensation) and for passengers. A single disruption can ripple through an aircraft's daily schedule because the same aircraft flies several legs. Aggregate delay rates alone do not show *where* that ripple is strongest.

This project is designed to help an operations analyst answer:

- Which delay causes account for the most lost minutes?
- Which airlines, airports, and routes show the weakest on-time performance?
- When an aircraft arrives late, how likely is its next departure to be late, and does turnaround time change that?

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3. Project Objectives and Analytical Questions

Objectives

1. Build a reproducible pipeline from raw flight records to an analysis-ready PostgreSQL layer.
2. Quantify delay volume by cause and airline.
3. Measure delay propagation between consecutive flights of the same aircraft.
4. Identify airport and route hotspots.

5. Deliver the results as an interactive dashboard usable by non-technical stakeholders.

Analytical questions

#	Question	Dashboard page
Q1	What do overall volumes, delay rates, cancellations, and diversions look like, and how do they vary by month and state?	1
Q2	Which causes and airlines contribute most to attributed delay minutes?	2
Q3	How does the previous flight's delay, combined with turnaround time, relate to the next flight's delay?	3
Q4	Which airports and routes show the highest delay rates, and is route volume related to delay rate?	4

4. Dataset Description and Scope

Item	Value
Reporting period	January–December 2025
Original flight records processed	**7,001,619**
Aircraft sequence records	**6,903,449**
Valid aircraft connections (propagation analysis)	**6,688,507**
Geography	U.S. domestic flights
Airlines in dashboard filters	14 carrier codes: AA, AS, B6, DL, F9, G4, HA, MQ, NK, OH, OO, UA, WN, YX

The original records support the operations, delay-cause, airport, and route analyses. The aircraft sequence records link each flight to the previous flight of the same aircraft and support the propagation analysis on Page 3.

5. Technology Stack and Architecture

Layer	Tool	Role
Processing / ETL	Python, Jupyter Notebook	Ingestion, cleaning, transformation
Storage and modelling	PostgreSQL	Tables, aggregations, analytical views
Analytics	SQL	KPI calculations and aggregations
Visualization	Microsoft Power BI	Four-page interactive dashboard

Pipeline: Raw 2025 flight data → **Python / Jupyter** (ingest, clean, transform) → **PostgreSQL** (tables, analytical views, SQL aggregations) → **Power BI** (4-page interactive dashboard)

6. Data Ingestion, Cleaning, Transformation, and ETL

The ETL was carried out in Python and Jupyter Notebook, with the results loaded into PostgreSQL.

- **7,001,619** original flight records were processed.
- An **aircraft sequence** dataset of **6,903,449** records was produced to link each flight to the previous flight of the same aircraft.
- **6,688,507** valid aircraft connections were used for propagation metrics.
- Diverted preceding flights were found to have **missing delay information**. This was identified during the propagation analysis and handled transparently at the visual level (see Section 11).

Analytical bands used in the propagation analysis

Dimension	Bands
Previous-flight delay	On Time / Early, 1–15, 16–30, 31–60, 61–120, 120+ min
Turnaround time	0–30, 31–60, 61–90, 91–120, 121–180, 180+ min

7. PostgreSQL Layer and Analytical Views

PostgreSQL serves as the storage, transformation, and aggregation layer. SQL views and aggregations supply each dashboard page:

Analytical need	Dashboard page
Overall operations, monthly and state summaries	1
Delay minutes by cause and airline	2
Aircraft-connection and propagation metrics	3
Airport summaries	3, 4
Route summaries	4

Airport and route summary views apply eligibility thresholds. They therefore do **not** necessarily represent every flight in the original dataset (see Page 4 and Section 11).

8. KPI Definitions and Calculation Methodology

Four metric families are used. Keeping them separate avoids a common source of misreading.

Metric family	What it measures	Unit	Where used
Total attributed delay minutes	Sum of minutes assigned to a cause (carrier, weather, NAS, security, late aircraft)	Minutes (volume)	Page 2
Departure delay rate	Delayed departures ÷ flights in scope	% of flights	Pages 1, 3, 4
Average delay minutes	Mean departure delay	Minutes per flight	Pages 1, 4
Delay propagation rate	Delayed next flights ÷ connections that followed a previous flight with a recorded delay	% of connections	Page 3

Why the distinction matters: a cause can dominate *minutes* without dominating *flight counts*. A high delay rate does not imply a long average delay. A propagation rate uses a different denominator (connections after a delay) than the overall next-flight delay rate (all valid connections).

KPI values

KPI	Value	Basis
Total flights	~7M (7,001,619 records processed)	Page 1
Departure delay rate	21.45%	Page 1
Average departure delay	13.56	Page 1
Cancellation rate	1.47%	Page 1
Diversion rate	0.28%	Page 1
Total attributed delay	111,934,625 min (111.93M)	Sum of the five cause totals
Valid aircraft connections	6,688,507	Page 3
Delayed next flights	1,417,740	Page 3
Next-flight delay rate	21.20%	1,417,740 ÷ 6,688,507
Connections after a previous delay	2,469,518	Page 3
Propagated delays	983,355	Page 3
Propagation rate	≈39.82%	983,355 ÷ 2,469,518
Airport departure delay rate	21.94%	Page 4, eligible airports
Airports analyzed	102	Page 4
Flights analyzed	6.33M	Page 4, eligible airports
High-value routes analyzed	160	Page 4
Weighted average departure delay	13.48	Page 4

9. Dashboard Design: The Four Pages

All pages use a dark, KPI-card-first layout with slicers for airline (and month or previous-delay band where relevant).

Page	Title	Purpose
1	Flight Operations Overview	Headline operational health
2	Delay Root Causes	Where attributed delay minutes come from
3	Delay Propagation Analysis	How earlier delays relate to later ones
4	Airport & Route Performance	Hotspots by airport and route

Page 1: Flight Operations Overview

- **KPIs:** Total flights (7M), departure delay rate (21.45%), average departure delay (13.56), cancellation rate (1.47%), diversion rate (0.28%).
- **Visuals:** monthly departure delay rate (area/line), departure delay rate by state (map), delay cause contribution (donut), flight recovery outcomes (donut).
- **Slicers:** month, airline.
- **Delay cause shares shown:** late aircraft 39.19%, carrier 32.45%, NAS 21.87%, weather 6.35%.

Page 2: Delay Root Causes

- **KPIs:** Total attributed delay (111.93M), late aircraft (43.86M), carrier (36.33M), NAS (24.48M), weather (7.11M).
- **Visuals:** 100%-stacked delay-cause composition by airline; total delay minutes by airline (stacked).
- **Key operational insight shown on the page:** late-aircraft and carrier delays together account for 71.64% of attributed

minutes.

Cause	Attributed minutes	Share
Late aircraft	43,862,971	39.19%
Carrier	36,326,377	32.45%
NAS	24,483,761	21.87%
Weather	7,108,714	6.35%
Security	152,802	0.14%
Total	**111,934,625**	**100%**

Page 3: Delay Propagation Analysis

- **KPIs:** 6.69M total connections, 21.20% next-flight delay rate, 2.47M connections after a previous delay, and the propagation rate (≈39.82%).
- **Visuals:**
- Heatmap matrix: next-flight delay rate by previous-delay band × turnaround-time band.
- Top high-risk propagation segments (airline × previous delay × turnaround, with connection counts).
- Propagation rate by airline.
- Top 10 airports by departure delay rate.
- Next-flight delay rate by previous-flight delay band.
- **Slicers:** airline, previous-flight delay band.

KPI population versus visual exclusions. The diverted-flight category described in Section 11 was excluded at the **visual level** from three Page 3 visuals. The KPI cards were not recalculated after that exclusion; they describe the overall verified population, while the affected visuals describe a narrower, filtered population.

Page 4: Airport & Route Performance

- **KPIs (5):** airport departure delay rate 21.94%, 102 airports analyzed, 6.33M flights analyzed, 160 high-value routes analyzed, weighted average departure delay 13.48.
- **Visuals (5):** top 10 airports by departure delay rate; top 10 routes by departure delay rate; top 10 routes by flight volume; top 10 airports by average departure delay; scatter plot of route delay rate vs. flight volume.
- **Scope:** Page 4 covers 6.33M of the ~7.0M records. Airport and route views apply eligibility thresholds, so figures are not a census of every flight.

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10. Key Findings and Operational Insights

Findings are descriptive and associational. Values read from dashboard charts are approximate.

10.1 Delay volume and causes

Finding	Evidence
Late-aircraft and carrier causes dominate attributed minutes	71.64% combined (80,189,348 of 111,934,625 min)
Weather and security are small contributors to attributed minutes	6.35% and 0.14%
Cause mix differs by airline	Page 2 shows visibly different compositions across carriers; some carriers show a larger carrier-delay share, others a larger NAS share
Absolute minutes reflect volume as well as performance	The airline totals chart is not normalized by flight count and should not be read as a performance ranking

10.2 Seasonality

The monthly delay-rate line shows a mid-year peak (roughly 27–28% around months 6–7), a trough in month 9 (roughly 16–17%), and a sharp rise again in December.

10.3 Delay propagation

Finding	Evidence
Baseline next-flight delay rate	21.20% across all valid connections
Delays are more likely after a delayed predecessor	≈39.82% in the 2,469,518 connections following a recorded previous delay
Next-flight delay rate rises with the size of the previous delay	Roughly 10% after on-time/early arrivals, ~20% after 1–15 min, ~44% after 16–30 min, ~60% after 31–60 min, ~63% after 61–120 min (~59% for 120+ min)
Short turnarounds combine with larger prior delays	Heatmap cells with prior delays of 31+ minutes and turnarounds of 0–90 minutes show the highest rates, many above 80%
Longer turnarounds are associated with lower rates	In the 180+ min column, rates are far lower across bands (roughly 14–21%)

Airline-level differences	Propagation rate by airline ranges from roughly 25% to 48%, with WN and F9 highest and HA lowest

The high-risk segment table reports rates near 99% for some airline × delay × turnaround combinations, shown with their connection counts (roughly 1,400 to 26,800 in the visible rows) so that volume can be judged alongside rate.

10.4 Airports and routes

Finding	Evidence
Highest airport departure delay rates	SFB 28.7%, DFW 28.6%, DAL 27.2%, PBI 26.4%, BWI 25.7%
Highest average departure delay	DFW 19.2, PBI 19.2, SFB 18.4, ORD 17.9, TYS 17.6
Highest route delay rates	ORD→EWR 33.5%, DFW→LAS 33.1%, DFW→PHX 31.8%, ATL→EWR 31.5%, DEN→SFO 31.5%
Highest-volume routes	ORD↔LGA (11.2K each way), SFO↔LAX (10.8K), HNL↔OGG (10.6K), JFK↔LAX (10.2K)
Route volume vs. delay rate	The scatter plot shows no obvious strong visual relationship; the highest-volume routes span a wide range of delay rates (roughly 9% to 27%). No statistical test was performed

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11. Data-Quality Considerations and Limitations

11.1 Diverted-flight missing-data issue

A category initially labelled **"No Previous Delay Data"** contained **13,979 connections** associated with **diverted preceding flights**, for which delay information was missing. This category showed a **72.88%** next-flight delay rate, with unusually high rates in some short-turnaround matrix cells.

Because these rates reflect missing or atypical data rather than a clean "no delay" scenario, treating the category like the others would have been misleading. It was therefore **excluded from three Page 3 visuals**.

- The exclusion is **visual-level**. It was not used to recalculate the Page 3 KPI cards.
- The verified KPI values (6,688,507 connections; 21.20%; 2,469,518; ≈39.82%) are aggregate findings **before** visual-level exclusions.

11.2 Eligibility thresholds on airport and route views

Page 4 (and the airport chart on Page 3) rely on summary views that apply eligibility thresholds. They cover 102 airports, 160 routes, and 6.33M flights, and so do not necessarily represent every flight in the 7,001,619-record dataset. The Page 4 airport delay rate (21.94%) differs from the Page 1 overall rate (21.45%) because the two cover different populations.

11.3 Association is not causation

The propagation analysis shows that delayed flights tend to be followed by delayed flights, particularly with short turnarounds. It **cannot** establish that the earlier delay caused the later one. Possible confounders include shared congestion at a hub, weather affecting several flights, crew or schedule structure, time of day, and airline scheduling practice. No causal model, statistical test, or controlled comparison was performed.

11.4 Other limitations

- Late-aircraft delay and the propagation metric are conceptually related but built differently; they should not be summed or treated as independent evidence.
- Single-year (2025) scope; no multi-year comparison.
- No predictive modelling, forecasting, or significance testing is included.
- Figures read from charts (seasonality, airline propagation ranges, heatmap ranges) are approximate.

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12. Recommendations for Airline Operations Analysis

These are **analytical suggestions** derived from observed associations, not proven causal interventions. Their impact has not been tested.

#	Suggestion	Rationale from the data
1	Prioritize investigation of late-aircraft and carrier-controllable delays	Together 71.64% of attributed minutes
2	Review turnaround buffers on segments with large prior delays	Highest next-flight delay rates occur with short turnarounds after prior delays of 31+ minutes
3	Examine airline-specific propagation patterns	Propagation rates differ materially across carriers; investigate scheduling structure before drawing conclusions
4	Study hotspot airports and routes in context	SFB, DFW, DAL, PBI and routes such as ORD→EWR and DFW→LAS show the highest rates; check congestion, time of day, and fleet mix
5	Normalize before ranking	Compare airlines using rates and per-flight metrics, not raw delay minutes

6	Extend the analysis	Add time-of-day, hub vs. non-hub, and statistical comparison in future work

13. Conclusion

This project turns 7,001,619 flight records into a four-page decision-support dashboard. It shows that delay minutes are concentrated in late-aircraft and carrier causes, that next-flight delay rates rise sharply following delayed predecessors (especially with short turnarounds), and that delay hotspots can be located at the airport and route level. The analysis is explicit about its boundaries: visual-level exclusion of the diverted-flight category, eligibility thresholds in summary views, and the distinction between association and causation.

14. Reproducibility Notes

The workflow follows the pipeline order below.

Step	Action
1	Run the Python / Jupyter ETL notebooks to ingest, clean, and transform the 2025 flight records
2	Load the processed data into PostgreSQL
3	Create the analytical views and SQL aggregations
4	Connect Power BI to the PostgreSQL views and refresh the dashboard
5	Validate against the verified totals: 7,001,619 records processed; 6,903,449 aircraft sequence records; 6,688,507 valid connections

Prepared as a portfolio project. Figures described as approximate were read from dashboard visuals; all other values are verified results from the analysis.